

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			Total area under graph = total distance travelled by bus (1) [Award on attempt to calculate total area under graph, not 25×190 ! : use of trapezium rule, or multiple areas] Substitution: $(0.5 \times 20 \times 25) + (25 \times 130) + (0.5 \times 40 \times 25)$ Or $0.5 \times (130 + 190) \times 25$ (1) = 4 000 [m] (1) Other distances between 3900 m and 4 100 m → 2 marks up to this point] $v = \frac{4\,000}{190}$ ecf = 21.1 [m/s] (1) [20.5 – 21.6 m/s from above] Horizontal line at 21 m/s ecf (1) Tolerance $\pm \frac{1}{2}$ square. Ignore any parts of the line with $t > 190$ s & don't penalise slightly shorter lines Alternative: Use of equations of motion: $x = \frac{(v+u)t}{2} + (vt) + \frac{(v+u)t}{2}$ (1) Substitution: $x = \frac{(25+0)}{2} \times 20 + (25 \times 130) + \frac{(0+25)}{2} \times 40$ (1) = 4 000 [m] (1) Other distances between 3900 m and 4 100 m → 2 marks up to this point] $v = \frac{4\,000}{190}$ ecf = 21.1 [m/s] (1) Horizontal line at 21 m/s ecf (1) Tolerance $\pm \frac{1}{2}$ square. Ignore any parts of the line with $t > 190$ s & don't penalise slightly shorter lines	1 1	1 1 1		5	4	
				Question 4 total	9	8	0	17	11	0